AthDGC at CIDL 2026

Civis Diachronic Linguistics · Athens · CIVIS BIP

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Diachronic Greek treebank · Indo-European cross-lingual alignment Athens Digital Glossa Chronos Research Network

HFRI Project No. 20577 · Greece 2.0 NRRP · CVL-CDSAML

Why this matters at CIDL

Diachronic linguistics needs:

- continuous coverage across periods
- aligned witnesses across languages
- uniform syntactic annotation
- machine-queryable argument structure

No single existing resource provides all four. AthDGC does.

Coverage (v0.4)

Period	Years	Annotated Greek rows
Archaic	8th–6th c. BC	Homer, Hesiod, Pindar, Sappho
Classical	5th–4th c. BC	Tragedy + comedy + history + philosophy + oratory
Koine	3rd c. BC – 4th c. AD	New Testament, Plutarch, Strabo, Lucian
Late Antique	4th–7th c. AD	Eusebius, Basil, Chrysostom, Procopius
Byzantine	7th–12th c. AD	Psellos, Anna Komnene, John of Damascus, Niketas
Late Byzantine	13th–15th c. AD	*in ingestion*
Early Modern	16th–18th c. AD	*in ingestion*

Cross-lingual alignment

NT verse-level cross-alignment to:

- **Latin** (Vulgate) — 6,861 verses aligned
- **Gothic** (Wulfila) — *in ingestion*
- **Old Church Slavonic** (Marianus) — *in ingestion*
- **Classical Armenian** — PROIEL-style pipeline in development

Method: LaBSE sentence embedding + AwesomeAlign word-level (mBERT attention).

Argument structure

For every VERB / AUX token AthDGC extracts:

- subject (`nsubj`, `nsubj:pass`, `csubj`)
- direct object (`obj`)
- indirect / oblique args (`iobj`, `obl:arg`, `obl:agent`, ...)
- voice (active / middle / passive)
- aspect (perfective / imperfective)
- tense, mood, verb-form

Stored per row in a column accessible to graph queries over the cross-lingual alignment edges.

Example query

> *"Show every Greek aorist transitive verb with an accusative object whose Latin Vulgate counterpart is a passive periphrastic"*

Translates to a Neo4j Cypher query traversing the `(Token)-[TRANSLATED_AS]->(Token)` edges where source carries `voice=Act,aspect=Perf,obj=Acc` and target carries `voice=Pass,verb_form=Periphrastic`.

Open infrastructure

- **Source code:** github.com/AthDGC/Diachronic-Linguistics-Platform (Apache-2.0)
- **Dataset DOI:** 10.5281/zenodo.20439182 (CC-BY-4.0)
- **Showcase:** athdgc.github.io
- **Hugging Face dataset:** `athdgc/diachronic-greek-corpus` (forthcoming)

Team

PI: Nikolaos Lavidas (EKPA) **International:** Dag Haug (Oslo, PROIEL Director), Leonid Kulikov (Ghent) **EKPA:** Kiki Nikiforidou, Vassiliki Geka, Vassileios Symeonidis, Theodoros Michalareas, Sofia Chionidi, Anastasia Tsiropina, Eleni Plakoutsi, Evangelos Argyropoulos

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Discussion

What CIDL participants can do with AthDGC today:

1. Query the corpus via the public Neo4j endpoint
2. Pull period-specific JSONL partitions from Hugging Face
3. Browse the PROIEL XML 2.0 samples on `athdgc.github.io`
4. Flag annotation errors via the showcase review toolbar
5. Cite version-pinned datasets via the version DOIs

Thank you

athdgc.github.io · github.com/AthDGC 10.5281/zenodo.20439182